

Regression Diagnostics and Assumptions

Residuals, normality, heteroscedasticity, influence and intervals explained clearly

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Why diagnostics matter

A model can produce a good metric and still be risky. Diagnostics check whether the model is behaving in a believable way. They help identify patterns in errors, unstable rows, assumption violations and statistical interpretation risks.

Residuals

A residual is actual value minus predicted value. If residuals show no pattern, the model is missing mostly random noise. If residuals curve or fan out, the model may be missing nonlinearity or unequal error spread.

QQ plot and normality

A QQ plot compares residuals with a theoretical normal distribution. Points close to the line mean the residual distribution is roughly normal. Strong curves or tail deviations warn that normality assumptions may not hold.

Homoscedasticity

Equal residual spread across prediction levels makes inference more trustworthy. Unequal spread is heteroscedasticity. It can make ordinary standard errors, p-values and confidence intervals less reliable.

Breusch-Pagan test

The Breusch-Pagan test is a formal test for heteroscedasticity. It does not replace visual inspection. It supports the decision about whether robust standard errors are needed.

Cook distance and leverage

Leverage checks unusual feature positions. Cook distance checks how much one row influences the fitted model. Rows with high influence should be investigated before finalizing business recommendations.

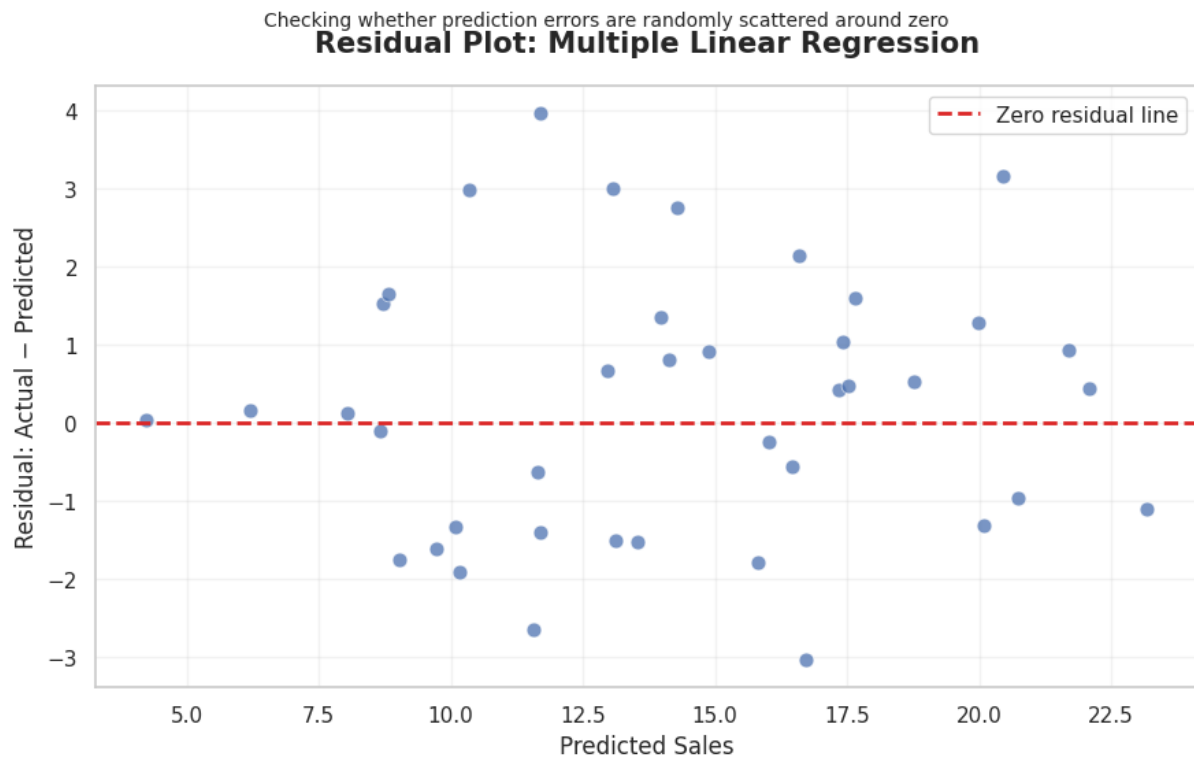
Robust standard errors

Robust standard errors adjust inference when residual spread is not constant. They do not change the prediction line, but they can change how confident we are about coefficient significance.

Prediction vs confidence intervals

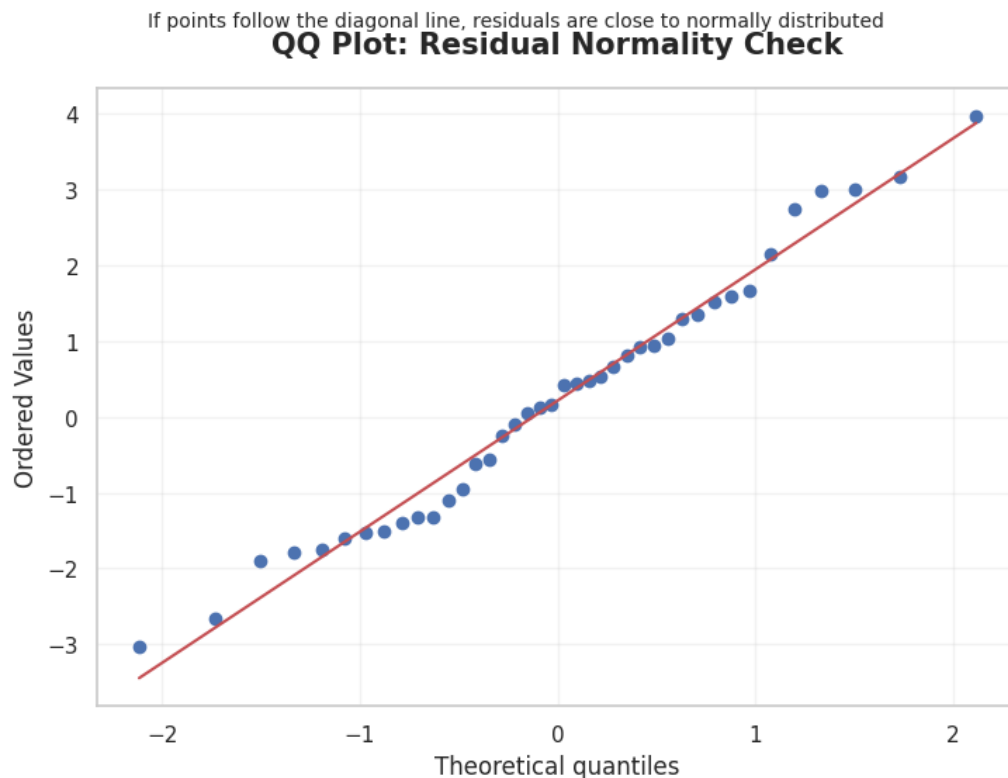
A confidence interval describes uncertainty around the average expected response. A prediction interval describes uncertainty around one future observation, so it is wider. This difference is important for forecasting decisions.

25. Residual Plot The residual plot checks whether prediction errors show a pattern



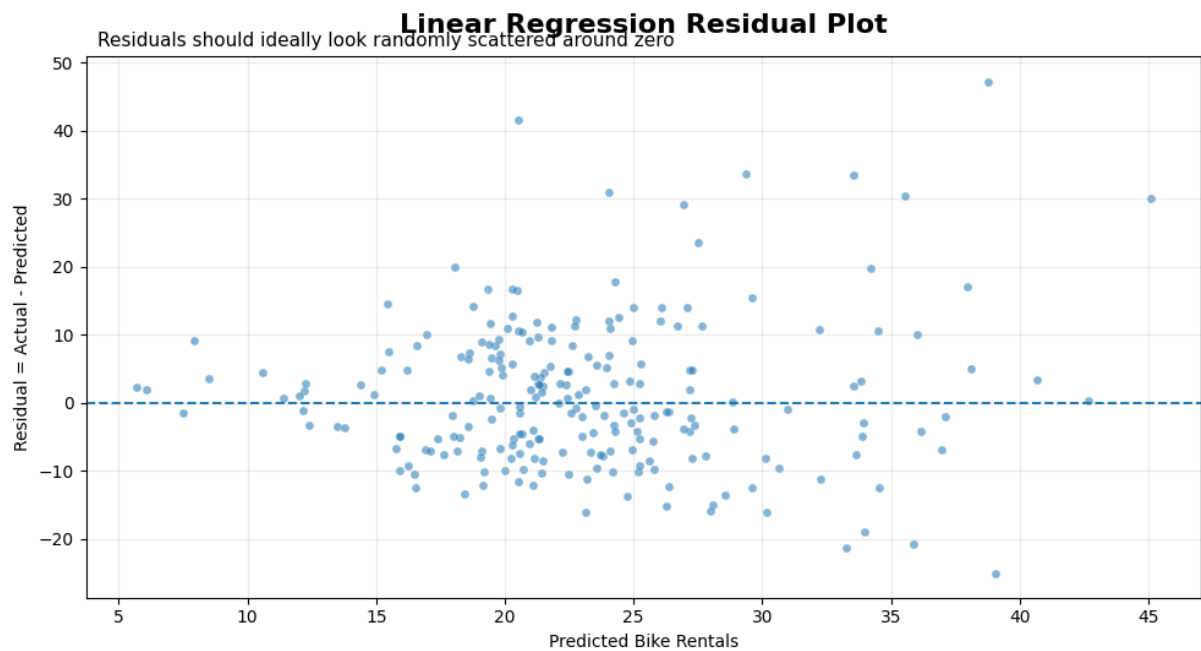
25. Residual Plot The residual plot checks whether prediction errors show a pattern. Good sign: residuals are randomly scattered around zero. Warning sign: curved pattern, funnel shape, or systematic trend. This helps check linearity and constant error spread.

26. QQ Plot for Residual Normality The QQ plot checks whether residuals are appr



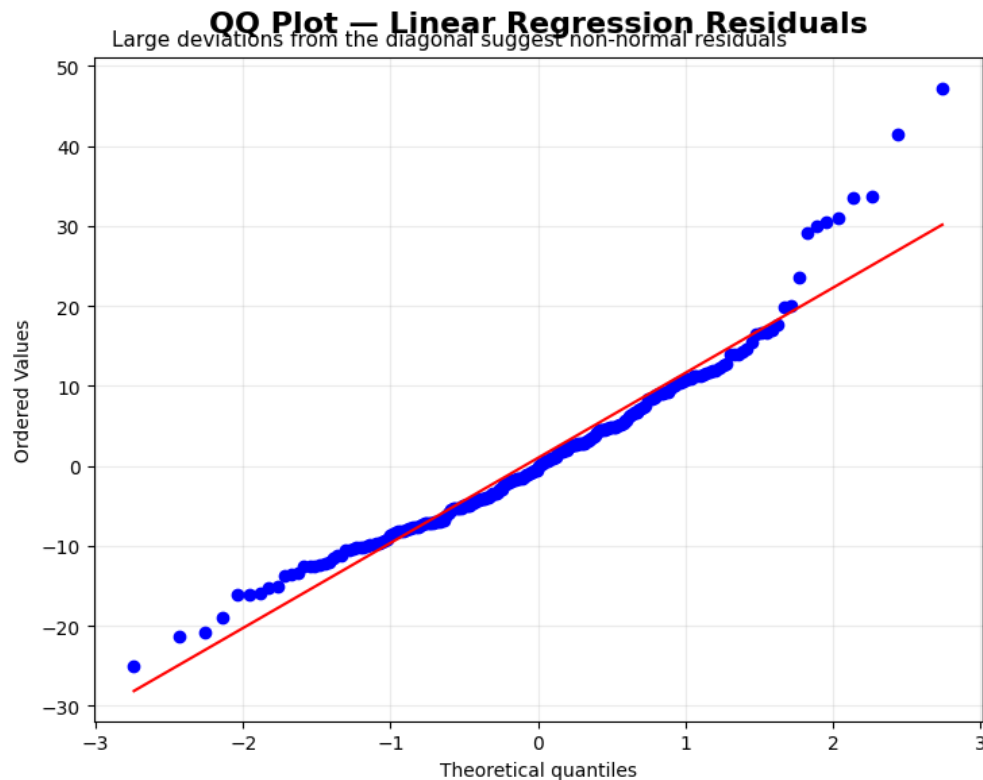
26. QQ Plot for Residual Normality The QQ plot checks whether residuals are approximately normally distributed. Simple explanation: if the points follow the diagonal line, residuals are close to normal. If the tails bend far away, the residual distribution may be skewed or heavy-tailed. Why it matters: normality is especially important for statistical inference such as p-values and confidence intervals.

14. Residual Plot for Linear Regression A residual plot shows whether the model



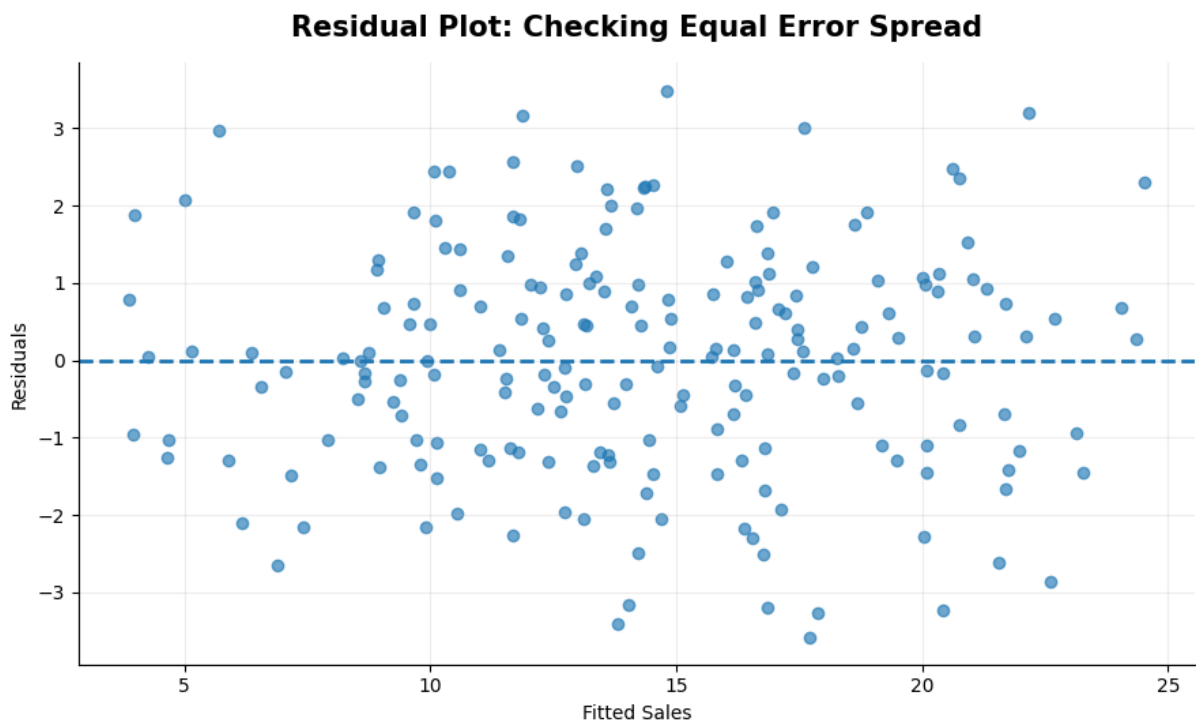
14. Residual Plot for Linear Regression A residual plot shows whether the model makes systematic errors. A good residual plot should look like random noise around zero. Patterns suggest that the model is missing structure. Business meaning: if the model underpredicts during high-demand periods, the business may underestimate staffing, inventory or bike availability needs.

15. QQ Plot for Residual Normality The MVD specifically asks us to check residuals



15. QQ Plot for Residual Normality The MVD specifically asks us to check residuals for normality using a QQ plot. A QQ plot compares model residuals with a normal distribution. If the points follow the diagonal line, residuals are roughly normal. Why it matters: count data often produces non-normal residuals. This gives us evidence for trying Poisson and Negative Binomial models.

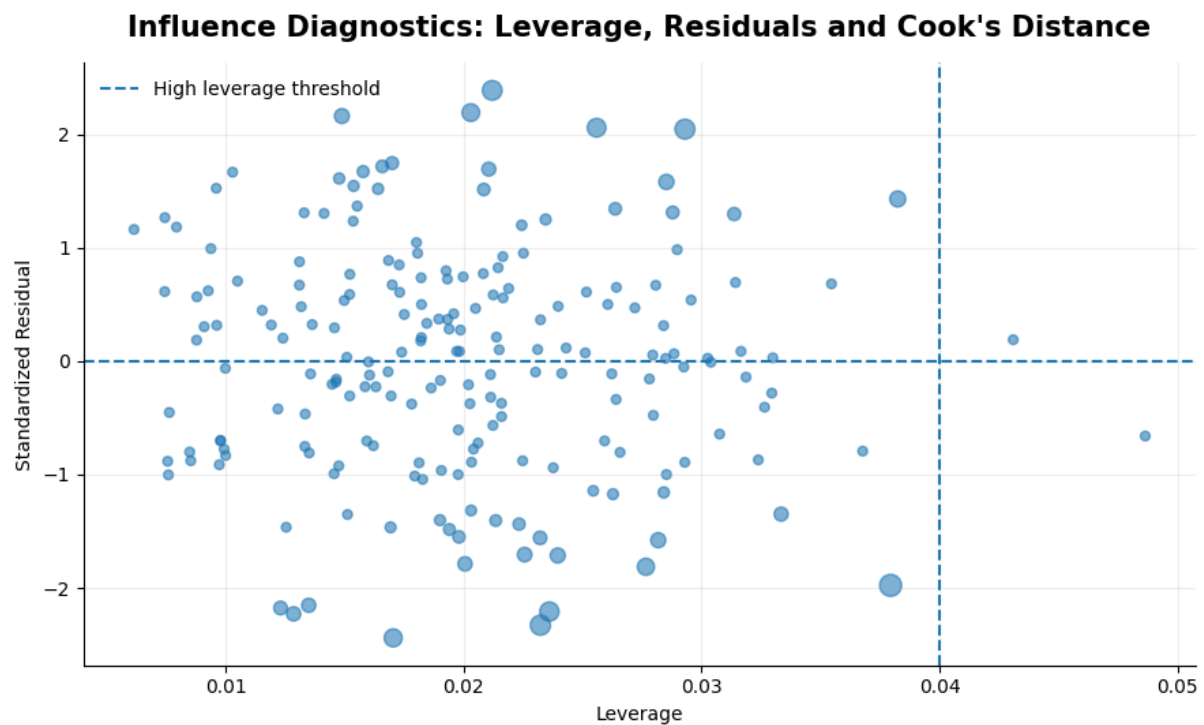
Code cell 12 — Homoscedasticity vs heteroscedasticity visual check Homoscedastic



Code cell 12 — Homoscedasticity vs heteroscedasticity visual check Homoscedasticity means residuals have roughly equal spread across prediction levels. Heteroscedasticity means the spread changes, often forming a funnel shape. This matters because unequal spread can make standard errors and p-values less trustworthy.

Code cell 17 — Cook's distance and leverage chart

This chart shows which rows de



Code cell 17 — Cook's distance and leverage chart This chart shows which rows deserve attention. Rows with high leverage and large residuals are often the most important to review.

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.